

Loan Approval: Financial Risk Analysis

A study of 20,000 loan applications: which applicant details are linked to a loan being approved, and which ones make little difference

Applications	Approved	Approval rate	Strongest factor	Model accuracy
20,000	4,780	23.9%	Annual income	88.3%

Tools: Python, pandas, seaborn, matplotlib and scikit-learn

Notebook, data and README:

github.com/nachiket-1/Loan-Approval-Financial-Risk-Analysis

This PDF is a readable copy of the whole project for phones and tablets. It follows the notebook from start to finish: the data, the checks, the findings for every variable, the prediction model and the conclusions. The code is in the notebook and the main parts are listed in Appendix B.

1. Overview

The goal of this project is to find out which applicant characteristics are linked to a loan being approved, and which are not. Every variable is studied the same way: applicants are split into groups, and the share of each group that gets approved (the approval rate) is compared.

Only about one application in four is approved (23.9%). Income decides most of it. The highest earning quarter of applicants is approved about 65% of the time, the lowest earning quarter about 1% of the time. The size of the loan compared with income, and the share of income already going to debt, matter almost as much. Savings balance and job tenure make almost no difference.

A logistic regression built only from applicant information predicts 88.3% of decisions correctly on applications it has never seen, against 76.1% for a model that rejects everyone.

2. Dataset and preparation

The data comes from the Financial Risk for Loan Approval dataset on Kaggle. The file `Loan.csv` has 20,000 applications and 36 columns, with no missing values and no duplicate rows.

The dataset is synthetic. The application dates run one per day from January 2018 to October 2072, so there is no real trend over time. The dates are only used in section 4.1 to confirm this, and not in the rest of the analysis.

Checks done on the data

1. Data types were checked. `ApplicationDate` was stored as text and was converted to a date type.
2. Missing values were counted: none.
3. Duplicate rows were counted: none.
4. Eleven columns of interest were kept in a separate copy of the data, so the original stays untouched.

Columns used

Column	Meaning
AnnualIncome	Yearly income of the applicant
CreditScore	Credit score (343 to 712 in this data)
EmploymentStatus	Employed, Self-Employed or Unemployed
LoanAmount	Amount requested
SavingsAccountBalance	Money in the savings account
TotalAssets, TotalLiabilities	What the applicant owns and owes
JobTenure	Years in the current job
HomeOwnershipStatus	Mortgage, Own, Rent or Other
LoanApproved	The result: 1 approved, 0 rejected
ApplicationDate	Date of the application (not used for findings)
TotalDebtToIncomeRatio, InterestRate, RiskScore	Added later for a closer look
LoanToIncome (new)	Loan amount divided by annual income
AssetLiabilityRatio (new)	Total assets divided by total liabilities

3. How approval was measured

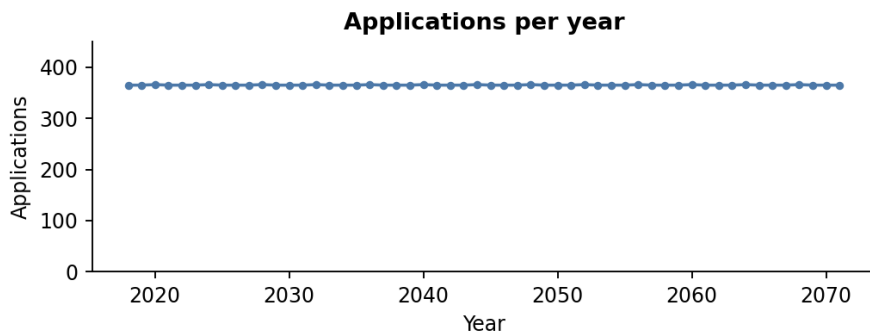
Counting how many approved loans each group has is misleading, because a bigger group always has more approvals. This project compares the **approval rate** instead: approved loans divided by applicants in the group.

Numeric variables are split into four groups of equal size (quartiles), so every group holds about 25% of the applicants. Credit score, the assets to liabilities ratio and job tenure use fixed bands that make sense for that variable. In every chart the red dashed line marks the overall approval rate of 23.9%. A group above the line is approved more often than average, a group below it less often.

4. Findings

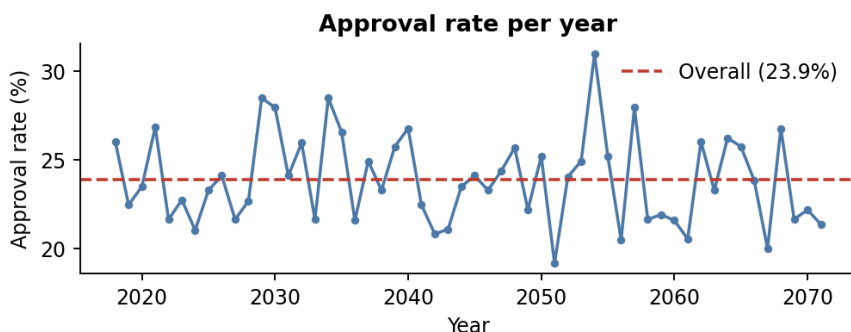
4.1 Applications over time

The data has exactly one application per day, so every full year has about 365 applications. The last year (2072) is incomplete and is left out of the charts.



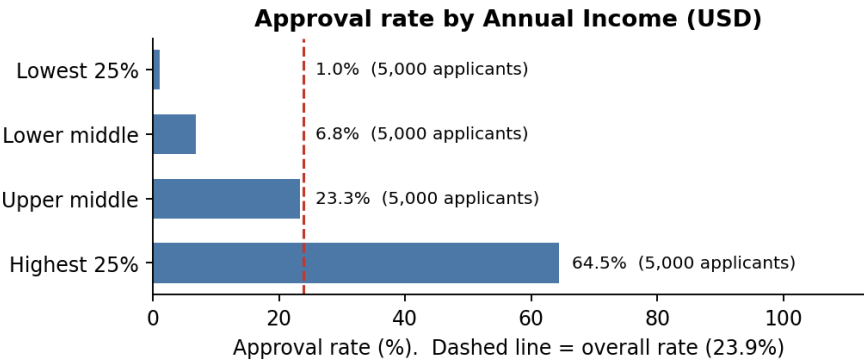
Applications per year. The line is flat because the dates are artificial.

The yearly approval rate moves between roughly 19% and 31% with no upward or downward trend, so time is not useful for explaining approval. It is dropped from the rest of the analysis.



Approval rate per year compared with the overall rate.

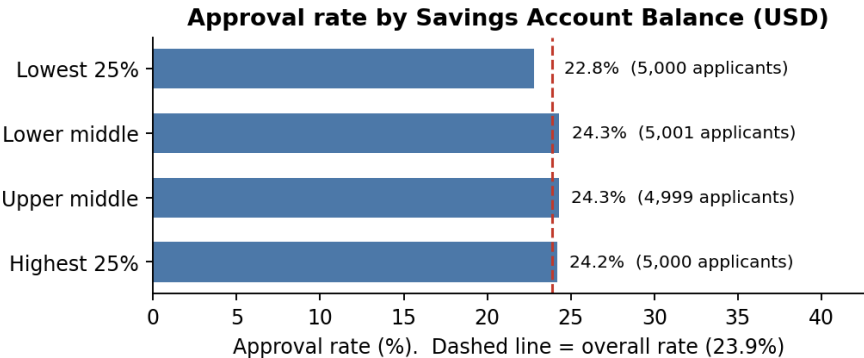
4.2 Annual income



Four equal income groups of 5,000 applicants each.

Income is the strongest applicant factor. The lowest quarter of earners (up to about USD 31,700 a year) is approved about 1% of the time, the next about 7%, the next about 23%, and the highest quarter (above about USD 74,400) about 65% of the time. The highest earners make up 67% of all approved loans (3,223 of 4,780).

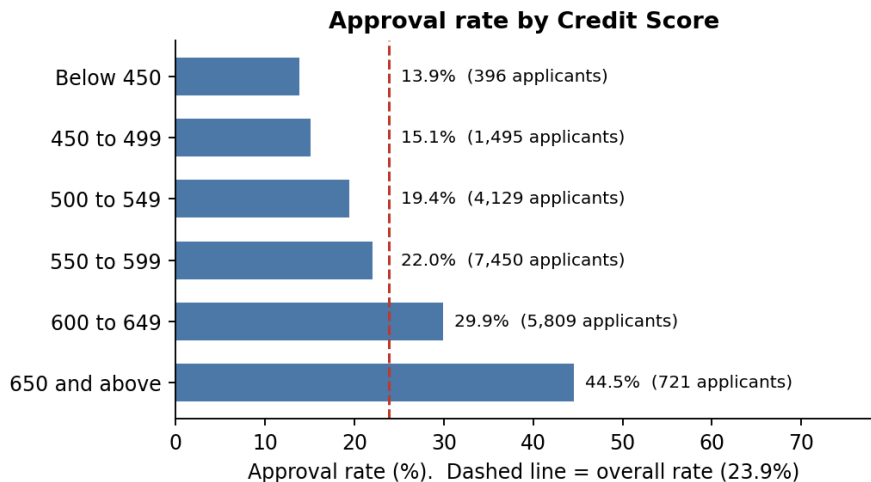
4.3 Savings account balance



Savings balance in four equal groups.

Savings balance has almost no effect. All four groups sit between 22.8% and 24.3%. The distribution is skewed: the average balance is about USD 4,950 but the median is about USD 2,990, because a few very large balances pull the average up.

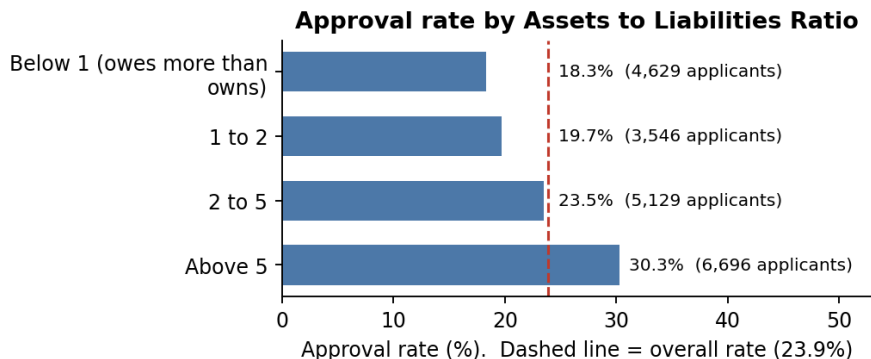
4.4 Credit score



Credit scores in 50 point bands.

Credit scores in this data run from 343 to 712, so the usual lender bands such as Very Good (740 and above) would be empty. Fifty point bands are used instead. The approval rate rises steadily, from about 14% below 450 to about 45% for 650 and above. Most applicants sit between 550 and 649 (13,259 of 20,000), so the best and worst bands are small. Credit score matters, but much less than income.

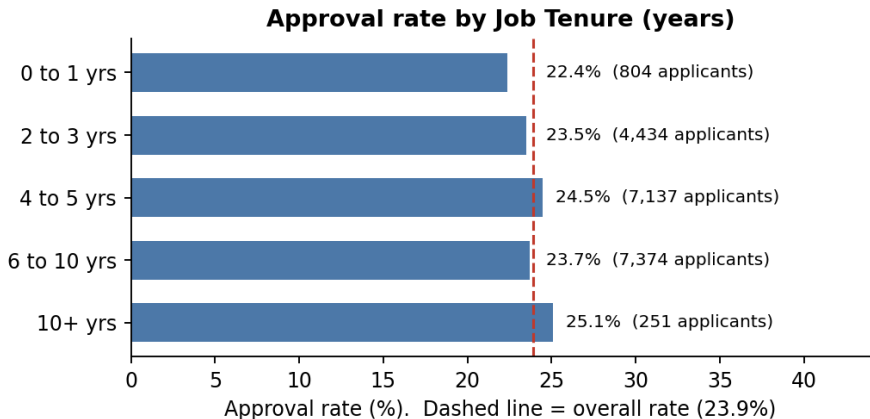
4.5 Assets to liabilities ratio



Total assets divided by total liabilities.

Applicants with more assets compared with what they owe are approved more often, from 18.3% when they owe more than they own to 30.3% when assets are more than five times liabilities. The ratio is very skewed (average about 7.2, median about 2.7, maximum above 550), so the median describes a typical applicant better. The effect is real but moderate.

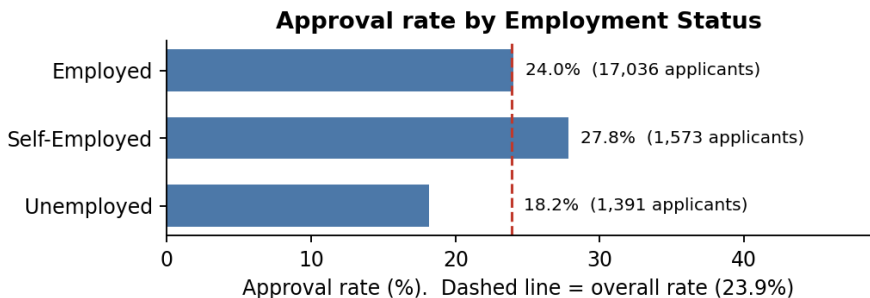
4.6 Job tenure



Years in the current job.

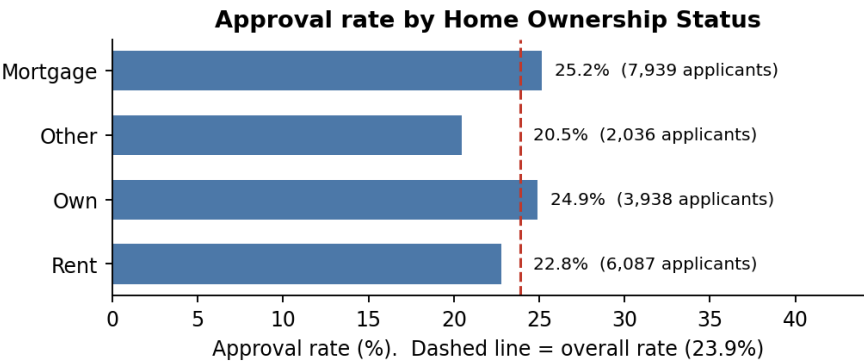
Job tenure makes almost no difference. Approval rates range only from 22.4% to 25.1%. The 0 to 1 year and 10+ year groups are small (804 and 251 applicants), so their rates are less reliable.

4.7 Employment status and home ownership



Approval rate by employment status.

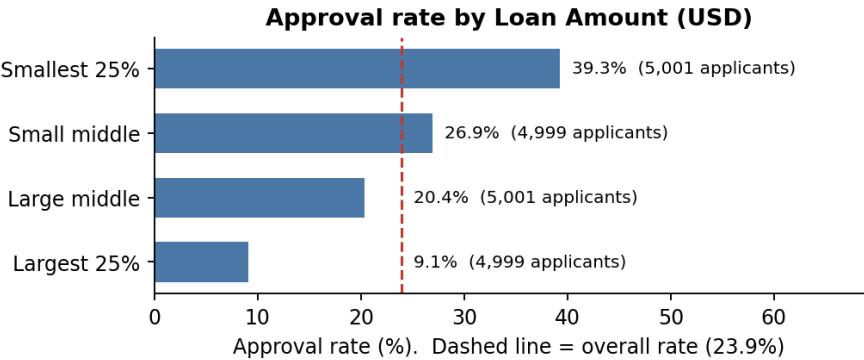
Unemployed applicants are approved least often (18.2%). Self-employed applicants are slightly above average (27.8%). Employed applicants are close to average (24.0%) and make up most of the data (17,036 applicants).



Approval rate by home ownership status.

Home ownership has a small effect. Applicants with a mortgage (25.2%) or who own their home (24.9%) are approved slightly more often than renters (22.8%) and the Other group (20.5%).

4.8 Loan amount

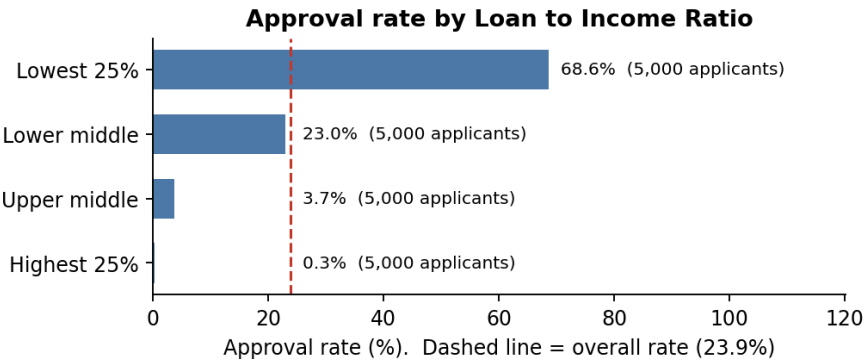


Loan amount in four equal groups.

The bigger the loan, the lower the approval rate: 39.3% for the smallest quarter of loans down to 9.1% for the largest quarter. Loan amount is the second strongest variable after income, and it works in the opposite direction.

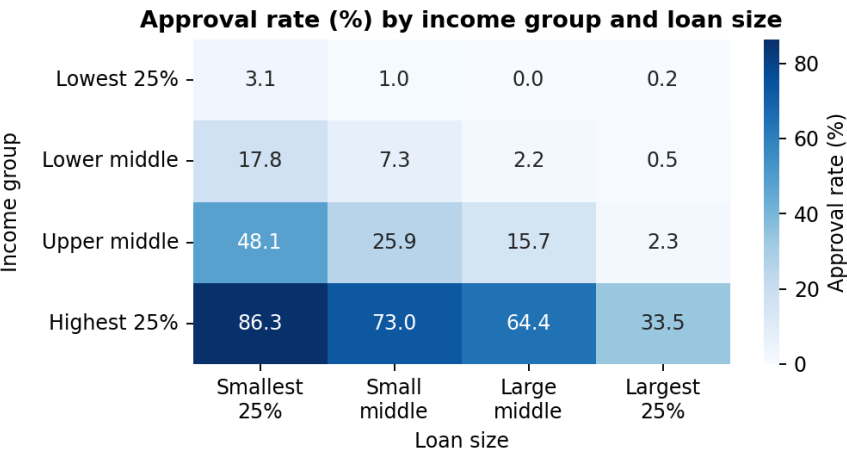
4.9 Loan amount compared with income

A loan of USD 30,000 is easy for a high earner and hard for a low earner, so a new column was created: loan amount divided by annual income. A value of 0.5 means the loan is half of the yearly income.



Loan to income ratio in four equal groups.

The ratio separates applicants very sharply. When the loan is up to about 26% of yearly income, about 69% of applications are approved. When it is above about 78% of yearly income, only 0.3% are (16 out of 5,000).

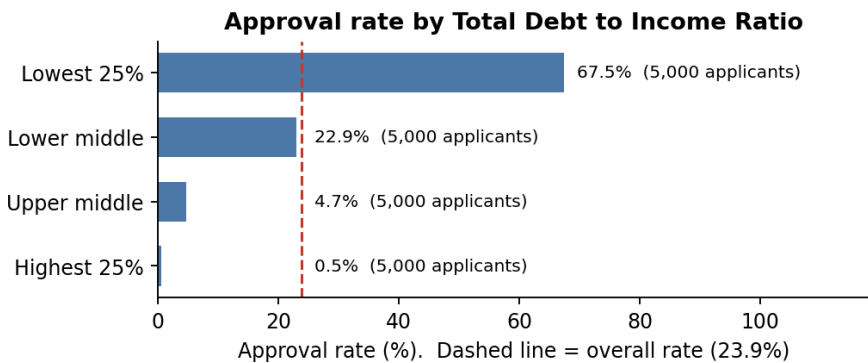


Approval rate for every combination of income group and loan size.

Low earners are almost never approved, even for the smallest loans (about 3%). High earners are approved 86% of the time for small loans but only 34% of the time for the largest ones. Loan size matters most for people who earn enough to be considered at all.

One correction to an early guess: the ratio does not beat income on its own in a simple correlation (about -0.41 against about 0.60), because it is very skewed and the link is not a straight line. It is still a much sharper divider than loan amount alone.

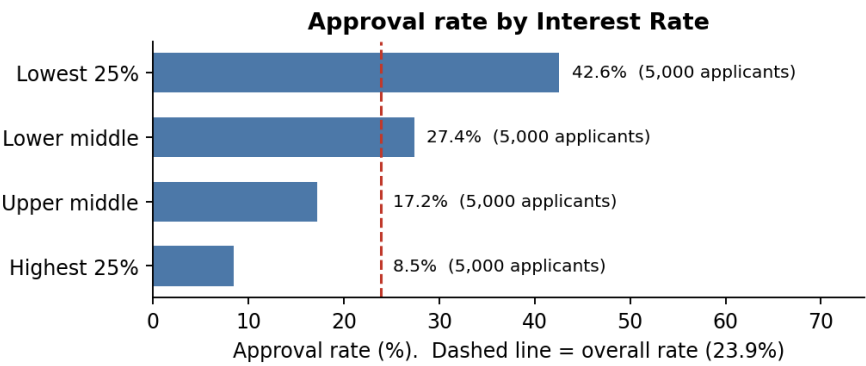
4.10 Debt to income ratio



Total debt to income ratio in four equal groups.

The less of their income an applicant owes, the more likely approval is. The lowest quarter (ratio up to about 0.18) is approved 67.5% of the time and the highest quarter (above about 0.51) only 0.5% of the time. This is a genuine applicant characteristic. It overlaps with income, which is expected because income is part of the ratio.

4.11 Interest rate

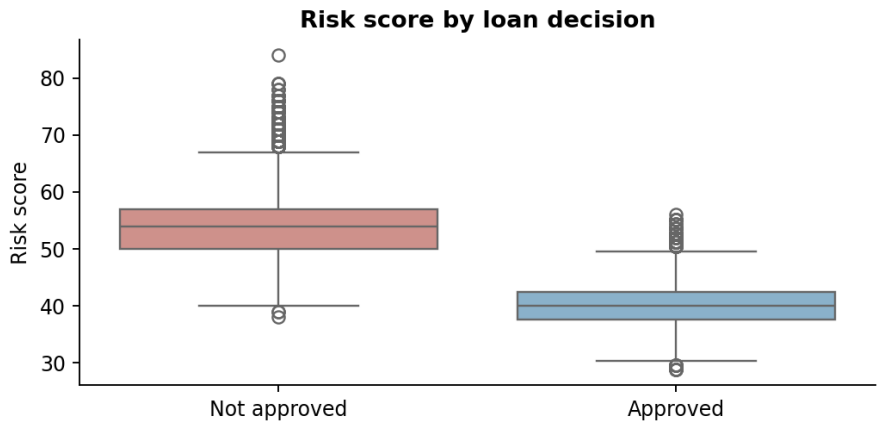


Interest rate in four equal groups.

Approval falls from 42.6% for the lowest interest rates to 8.5% for the highest. The interest rate is set by the lender and riskier applicants are usually offered higher rates, so it partly reflects the lender's view of the applicant. It was studied but kept out of the prediction model.

4.12 Risk score

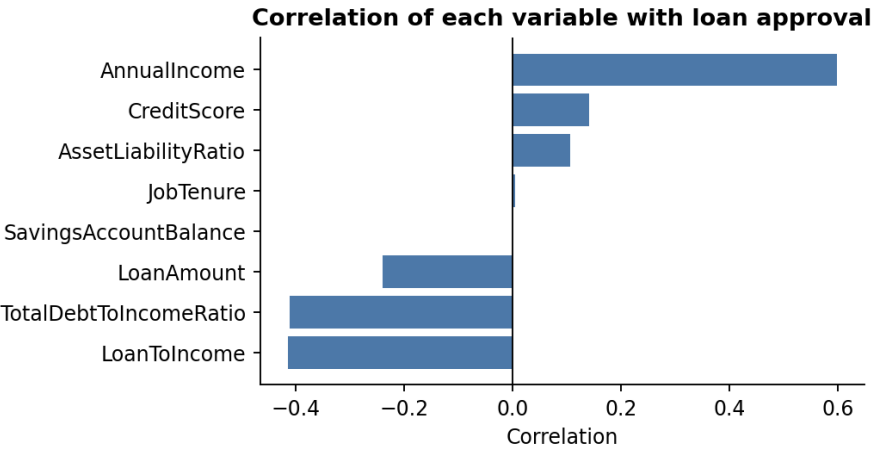
RiskScore has the strongest link to approval of any column (correlation about -0.77). Approved applicants have a clearly lower score (average about 40) than rejected ones (average about 54), and the two groups barely overlap.



Risk score for rejected and approved applications.

The score is most likely calculated from the same information the lender uses to make the decision. Using it to explain approval would be circular, like predicting exam results from the marks given by the examiner. It is therefore kept out of the main model.

4.13 All numeric variables together



Correlation with loan approval. Correlation shows association, not cause.

Variable	Lowest approval rate	Highest approval rate	Effect
Annual Income (USD)	Lowest 25% (1.0%)	Highest 25% (64.5%)	Very strong
Loan to Income Ratio	Highest 25% (0.3%)	Lowest 25% (68.6%)	Very strong
Total Debt to Income Ratio	Highest 25% (0.5%)	Lowest 25% (67.5%)	Very strong
Loan Amount (USD)	Largest 25% (9.1%)	Smallest 25% (39.3%)	Strong
Interest Rate	Highest 25% (8.5%)	Lowest 25% (42.6%)	Strong
Credit Score	Below 450 (13.9%)	650 and above (44.5%)	Moderate
Assets to Liabilities Ratio	Below 1 (owes more than owns) (18.3%)	Above 5 (30.3%)	Moderate
Employment Status	Unemployed (18.2%)	Self-Employed (27.8%)	Small
Home Ownership Status	Other (20.5%)	Mortgage (25.2%)	Small
Job Tenure (years)	0 to 1 yrs (22.4%)	10+ yrs (25.1%)	None
Savings Account Balance (USD)	Lowest 25% (22.8%)	Lower middle (24.3%)	None

5. Predicting approval

A logistic regression was trained to predict whether a loan is approved. It was chosen because it gives a probability of approval and its coefficients are easy to read.

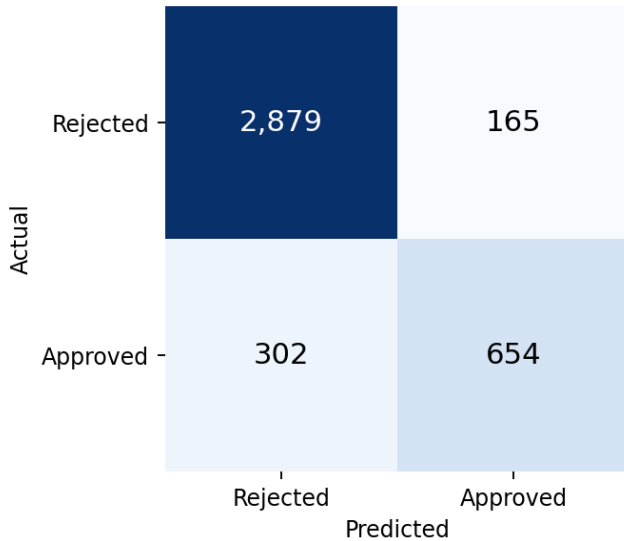
Features: annual income, credit score, loan amount, savings balance, job tenure, assets to liabilities ratio, loan to income ratio, debt to income ratio, employment status and home ownership. RiskScore and InterestRate were left out because the lender sets them. The data was split 80% for training (16,000 applications) and 20% for testing (4,000 applications), keeping the same approval share in both parts. Variables were scaled so the coefficients can be compared.

Metric	Without RiskScore (used)	With RiskScore (comparison only)
Accuracy	88.3%	98.3%
Precision	79.9%	97.1%
Recall	68.4%	95.7%
F1 score	73.7%	96.4%
ROC AUC	0.941	0.998

The model is right 88.3% of the time on applications it has never seen. A model that rejects every application would be right 76.1% of the time. Accuracy on the training data was 88.0%, almost the same, so the model is not overfitting. The ROC AUC of 0.94 means it ranks approved applicants above rejected ones very reliably.

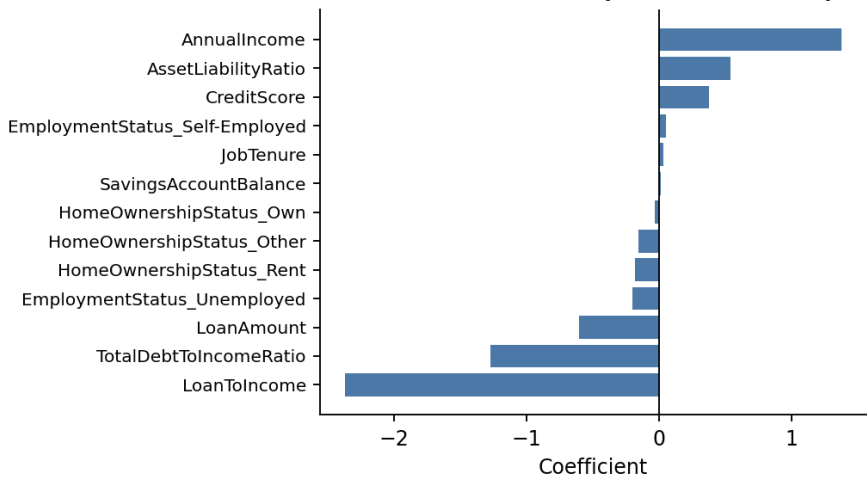
Of the loans the model predicts as approved, 79.9% really were (precision). Of the loans that really were approved, it finds 68.4% (recall). In the confusion matrix, 302 approved loans were missed and 165 rejected loans were wrongly predicted as approved.

Confusion matrix (4,000 test applications)



Confusion matrix on the test data.

Model coefficients (scaled variables)



Positive coefficients push towards approval, negative towards rejection.

The loan to income ratio is the strongest push towards rejection, followed by the debt to income ratio and the loan amount. Annual income, the assets to

liabilities ratio and credit score push towards approval. Savings balance and job tenure are close to zero, which matches the earlier findings. Income, loan amount and loan to income overlap, so the overall pattern matters more than any single coefficient.

Choosing the cutoff

The default cutoff is 0.5. Lowering it finds more approved loans but is less precise. The best choice depends on the cost of a bad loan compared with a lost customer.

Cutoff	Accuracy	Precision	Recall
0.3	87.1%	68.7%	84.1%
0.4	88.2%	74.6%	76.9%
0.5	88.3%	79.9%	68.4%

What happens with RiskScore

Adding RiskScore lifts accuracy from 88.3% to 98.3% and the ROC AUC from 0.94 to 0.998. That looks impressive but it is the circular effect described earlier: the score already contains the lender's judgement, so the model is partly copying the answer. The honest model is the one without RiskScore, and 88% accuracy from applicant information alone is a good result.

6. Improvements made along the way

The first version of the analysis was reviewed and corrected in these ways.

1. **A typo changed the meaning of a note.** The note under the approved applicants chart said the lowest earners had the "list approved" applications, which reads as the most approved. The chart shows the opposite: the highest earners have the most approved loans and the lowest earners the fewest. The note was corrected.
2. **Income groups were uneven.** Using the mean as a cutoff made one group much wider than the others (7,522 applicants against 5,000 or fewer). Equal sized quartiles replaced it.
3. **Counts were replaced by approval rates**, since counts mostly show where the most applicants are.

4. **A pandas warning** (SettingWithCopyWarning) was removed by working on a proper copy of the data.
5. **The time chart** was corrected for the incomplete last year, and a note explains that the dates are artificial.
6. **Repeated code** was moved into three small helper functions, and the mode now prints as a single number.

7. Conclusions and next steps

Conclusions

1. Annual income is the strongest applicant factor. Approval climbs from about 1% in the lowest income quarter to about 65% in the highest.
2. What matters is the size of the loan compared with income. Approval is about 69% when the loan is up to a quarter of yearly income and almost 0% when it is above three quarters of it. Debt to income ratio tells the same story (67.5% against 0.5%).
3. Loan amount works against the applicant (39% for the smallest loans, 9% for the largest), but low earners are rejected even for small loans.
4. Credit score, assets to liabilities ratio, employment status and home ownership have a moderate or small effect.
5. Savings balance and job tenure have almost no effect, and application date shows no trend.
6. Interest rate and RiskScore are set by the lender, so they were not used to explain approval. A logistic regression with applicant information only is right 88.3% of the time on test data (baseline 76.1%).
7. Most applications (76%) are rejected.

What to do next

1. Try a tree based model such as a random forest, because approval looks like a threshold effect.
2. Use cross validation so the result does not depend on one train and test split.
3. Choose the cutoff from the real cost of a bad loan against a lost customer instead of using 0.5.
4. Add the other applicant columns (age, education, bankruptcy history, previous defaults) and check whether the model improves.

Appendix A. Approval rate tables

Every group for every variable. Approval rate is approved loans divided by applicants.

Annual Income (USD)

Group	Range of values	Applicants	Approved	Rate
Lowest 25%	15,000 to 31,676	5,000	52	1.0%
Lower middle	31,680 to 48,565	5,000	342	6.8%
Upper middle	48,567 to 74,388	5,000	1,163	23.3%
Highest 25%	74,400 to 485,341	5,000	3,223	64.5%

Loan to Income Ratio

Group	Range of values	Applicants	Approved	Rate
Lowest 25%	0.03 to 0.26	5,000	3,431	68.6%
Lower middle	0.26 to 0.45	5,000	1,149	23.0%
Upper middle	0.45 to 0.78	5,000	184	3.7%
Highest 25%	0.78 to 7.54	5,000	16	0.3%

Total Debt to Income Ratio

Group	Range of values	Applicants	Approved	Rate
Lowest 25%	0.02 to 0.18	5,000	3,374	67.5%
Lower middle	0.18 to 0.30	5,000	1,146	22.9%
Upper middle	0.30 to 0.51	5,000	236	4.7%
Highest 25%	0.51 to 4.65	5,000	24	0.5%

Loan Amount (USD)

Group	Range of values	Applicants	Approved	Rate
Smallest 25%	3,674 to 15,575	5,001	1,964	39.3%
Small middle	15,578 to 21,914	4,999	1,345	26.9%
Large middle	21,915 to 30,835	5,001	1,018	20.4%
Largest 25%	30,836 to 184,732	4,999	453	9.1%

Interest Rate

Group	Range of values	Applicants	Approved	Rate
Lowest 25%	0.113 to 0.209	5,000	2,129	42.6%
Lower middle	0.209 to 0.235	5,000	1,368	27.4%
Upper middle	0.235 to 0.266	5,000	860	17.2%
Highest 25%	0.266 to 0.447	5,000	423	8.5%

Credit Score

Group	Range of values	Applicants	Approved	Rate
Below 450	343 to 449	396	55	13.9%
450 to 499	450 to 499	1,495	226	15.1%
500 to 549	500 to 549	4,129	801	19.4%
550 to 599	550 to 599	7,450	1,641	22.0%
600 to 649	600 to 649	5,809	1,736	29.9%
650 and above	650 to 712	721	321	44.5%

Assets to Liabilities Ratio

Group	Range of values	Applicants	Approved	Rate
Below 1 (owes more than owns)	0.01 to 1.00	4,629	849	18.3%
1 to 2	1.00 to 2.00	3,546	700	19.7%
2 to 5	2.00 to 5.00	5,129	1,204	23.5%
Above 5	5.00 to 559.49	6,696	2,027	30.3%

Employment Status

Group	Range of values	Applicants	Approved	Rate
Employed		17,036	4,089	24.0%
Self-Employed		1,573	438	27.8%
Unemployed		1,391	253	18.2%

Home Ownership Status

Group	Range of values	Applicants	Approved	Rate
Mortgage		7,939	1,997	25.2%
Other		2,036	417	20.5%
Own		3,938	980	24.9%
Rent		6,087	1,386	22.8%

Job Tenure (years)

Group	Range of values	Applicants	Approved	Rate
0 to 1 yrs	0 to 1	804	180	22.4%
2 to 3 yrs	2 to 3	4,434	1,041	23.5%
4 to 5 yrs	4 to 5	7,137	1,748	24.5%
6 to 10 yrs	6 to 10	7,374	1,748	23.7%
10+ yrs	11 to 16	251	63	25.1%

Savings Account Balance (USD)

Group	Range of values	Applicants	Approved	Rate
Lowest 25%	73 to 1,541	5,000	1,140	22.8%
Lower middle	1,542 to 2,986	5,001	1,216	24.3%
Upper middle	2,987 to 5,873	4,999	1,214	24.3%
Highest 25%	5,874 to 200,089	5,000	1,210	24.2%

Appendix B. Core code

The main steps of the notebook. The full code, with every chart, is in `Loan_Approval_Financial_Risk_Analysis.ipynb`.

Load, clean and select columns

```
loans_data = pd.read_csv('Loan.csv')
loans_data['ApplicationDate'] = pd.to_datetime(loans_data['ApplicationDate'])

columns_ = ['ApplicationDate', 'AnnualIncome', 'CreditScore',
            'EmploymentStatus', 'LoanAmount', 'SavingsAccountBalance',
            'TotalAssets', 'TotalLiabilities', 'JobTenure',
            'HomeOwnershipStatus', 'LoanApproved']
loans = loans_data[columns_].copy()
```

Approval rate for each group

```
def approval_summary(data, category_col):
    summary = data.groupby(category_col, observed=True)['LoanApproved'].agg(
        Applicants='size', Approved='sum', ApprovalRate='mean')
    summary['ApprovalRate'] = (summary['ApprovalRate'] * 100).round(1)
    return summary

loans['Income Category'] = pd.qcut(loans['AnnualIncome'], q=4,
    labels=['Lowest 25%', 'Lower-middle', 'Upper-middle', 'Highest 25%'])
print(approval_summary(loans, 'Income Category'))
```

New columns

```
loans['LoanToIncome'] = loans['LoanAmount'] / loans['AnnualIncome']

liabilities = loans['TotalLiabilities'].replace(0, np.nan)
loans['AssetLiabilityRatio'] = loans['TotalAssets'] / liabilities
```

Logistic regression

```
X = pd.get_dummies(loans[numeric_features + categorical_features],
    columns=categorical_features, drop_first=True)
y = loans['LoanApproved']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

model = Pipeline([('scale', StandardScaler()),
    ('logreg', LogisticRegression(max_iter=1000))])
model.fit(X_train, y_train)
```